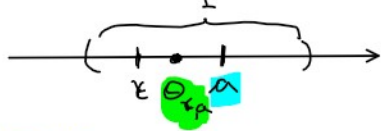


Formula lui Taylor

$I = \text{interval deschis}$, $C^k(I) = \{f: I \rightarrow \mathbb{R} : f^{(i)}, i=0, \dots, k; f^{(k)} \text{ cont.}\}$

T₁ (formula lui Taylor cu restul Lagrange)

$f \in C^{n+1}(I)$, $a \in I \Rightarrow f(x) = \underbrace{T_n f(x)}_{\text{pol. Taylor de ord. } n \text{ în jurul lui } a} + \underbrace{R_n f(x)}_{\text{rest}}$, $x \in I$, unde



$$T_n f(x) = f(a) + \frac{f'(a)}{1!} \cdot (x-a) + \dots + \frac{f^{(n)}(a)}{n!} \cdot (x-a)^n$$

$$R_n f(x) = \frac{f^{(n+1)}(\theta_{x,a})}{(n+1)!} \cdot (x-a)^{n+1}, \quad \theta_{x,a} = \text{pt. intermediară între } x \text{ și } a$$

$T_n f(x) = \text{taylor}(f, x, a, \text{'order'}, n+1) \rightarrow \text{apel în Octave}$
 $\gg \text{help @sym/taylor}$

Serii de puteri MacLaurin ($a=0$)

$$e^x = 1 + \frac{x^1}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots, \quad x \in \mathbb{R}$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^m}{(2m+1)!} \cdot x^{2m+1} + \dots, \quad x \in \mathbb{R}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{(-1)^m}{(2m)!} \cdot x^{2m} + \dots, \quad x \in \mathbb{R}$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + \frac{(-1)^{n-1}}{n} \cdot x^n + \frac{(-1)^n}{n+1} \cdot x^{n+1} + \dots, \quad |x| < 1$$

$$\frac{1}{1+x} = 1 - x + x^2 - \dots + (-1)^n \cdot x^n + \dots, \quad |x| < 1$$

$$\frac{1}{1-x} = 1 + x + x^2 + \dots + x^n + \dots, \quad |x| < 1$$

$$\frac{1}{(1+x)^k} = 1 + \sum_{n=0}^{\infty} \binom{k}{n} \cdot x^n, \quad |x| < 1$$

$$\frac{1}{(1+x)^{-\frac{1}{2}}} = \frac{1}{\sqrt{1+x}}$$

T₂ (crit. lui Leibniz pt. serii alternante)

$S = \sum_{k=0}^{\infty} (-1)^k a_k$, unde $(a_k)_{k \in \mathbb{N}}$ desc. cu $\lim_{k \rightarrow \infty} a_k = 0$ ($a_k \geq 0, k \rightarrow \infty$)
 conv. $S_n = \sum_{k=0}^n (-1)^k a_k \Rightarrow |S - S_n| \leq a_{n+1}$

$$\text{conv.} \quad S_n = \sum_{k=0}^n (-1)^k a_k = 1 \quad |S_n - S_{n+1}| \leq a_{n+1}.$$

T3 (teor. lui Abel pt. seri)

$$f(x) = \sum_{k=0}^{\infty} a_k x^k, \quad |x| < r$$

$$\sum_{k=0}^{\infty} a_k r^k \text{ conv.} \Rightarrow \lim_{x \rightarrow r} f(x) = \sum_{k=0}^{\infty} a_k r^k.$$

